AWS Amazon Inspector Lab: Network Assessment

Summary:-

In this lab, **Amazon Inspector** was used to perform a **Network Assessment** on an Amazon EC2 instance. The assessment analyzed the instance's network exposure by identifying publicly accessible ports and reporting potential security risks based on AWS security best practices. The findings were reviewed according to their severity levels, including **High**, **Medium**, **Low**, and **Informational**..

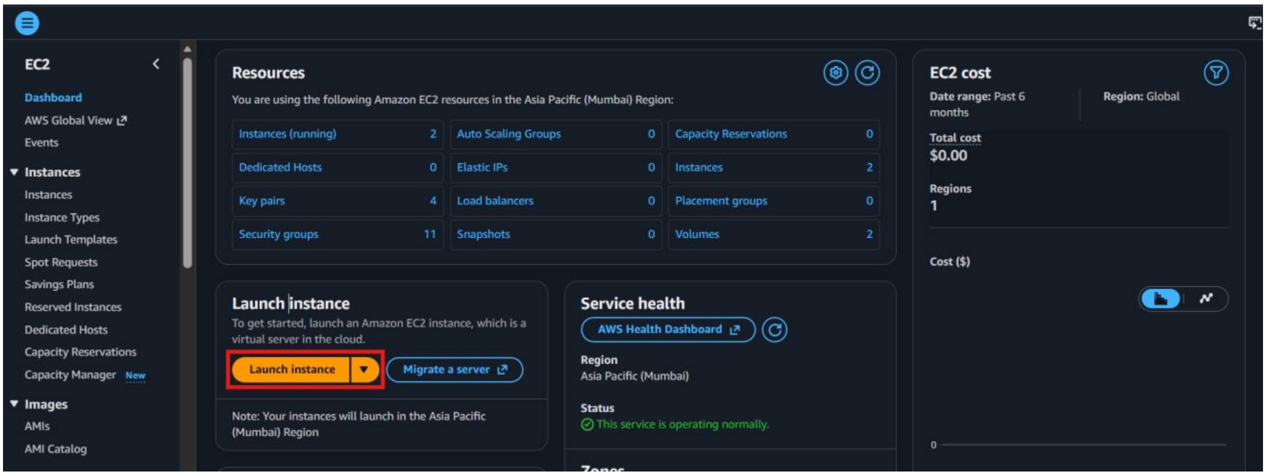
Let's Start:-

• Amazon Inspector is an automated security assessment service that helps identify security vulnerabilities and deviations from AWS security best practices.
• A Network Assessment checks which ports on an EC2 instance are reachable from outside the VPC.
• This assessment does not require installing the Amazon Inspector Agent.

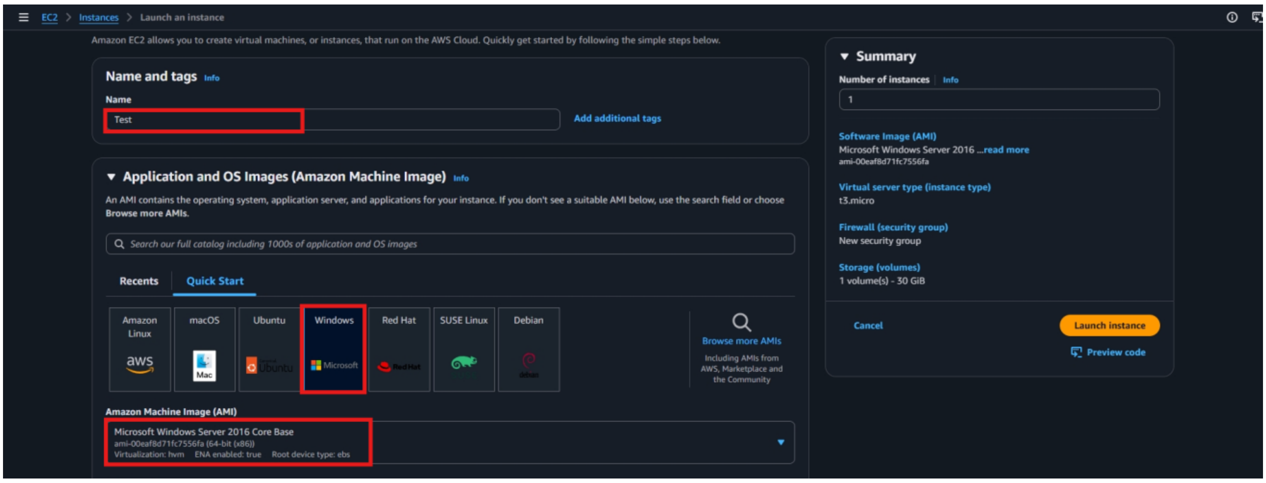
Lab Steps:-

Step 1 – Launch an EC2 Instance

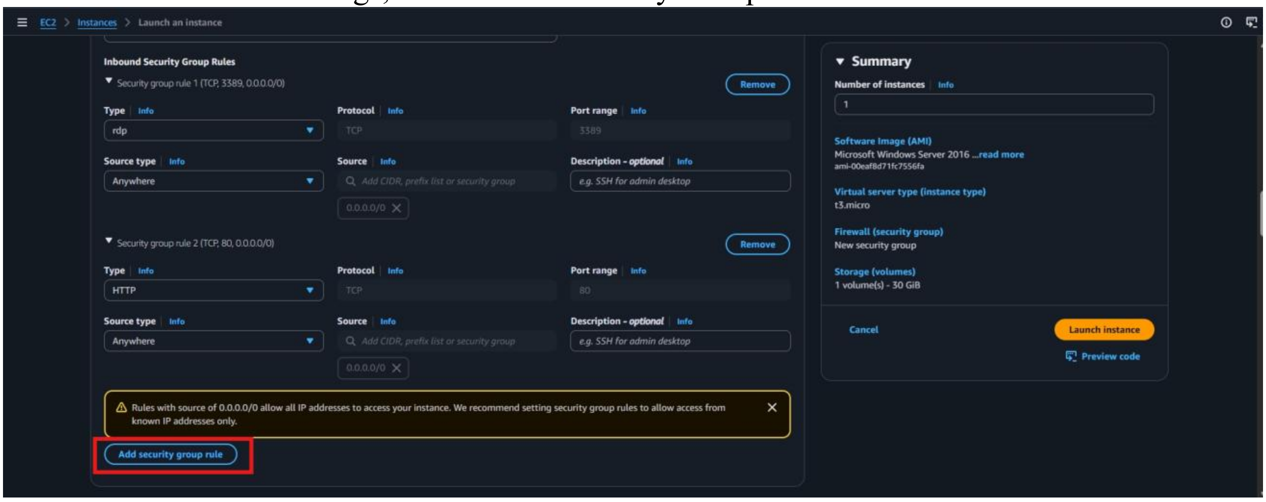
- Navigate to the Amazon EC2 console.
- Click Launch Instance.



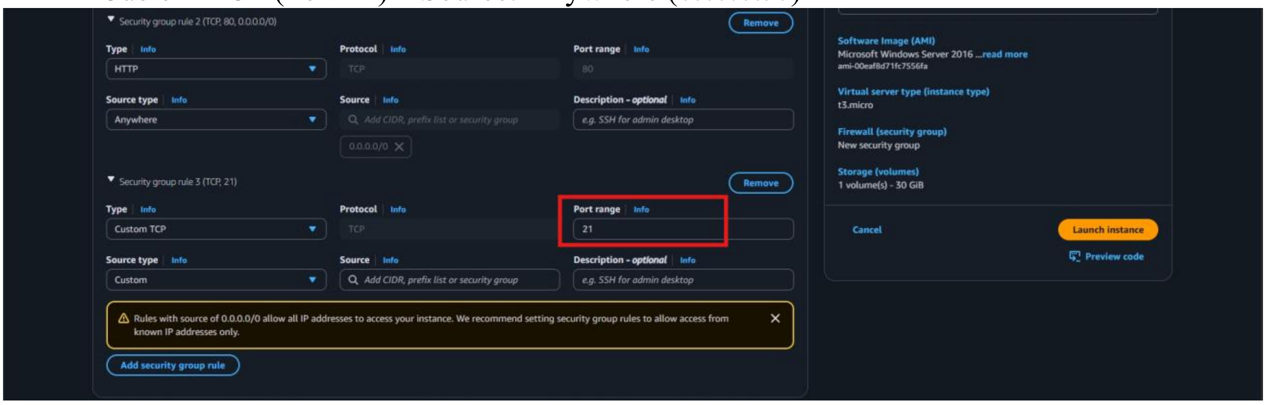
- Enter the instance name.
- Select Microsoft Windows Server 2016 Core Base as the Amazon Machine Image (AMI).



• Choose an eligible instance type (such as t2.micro, if available).
• Select or create a key pair.
• Under Network Settings, create a new Security Group.



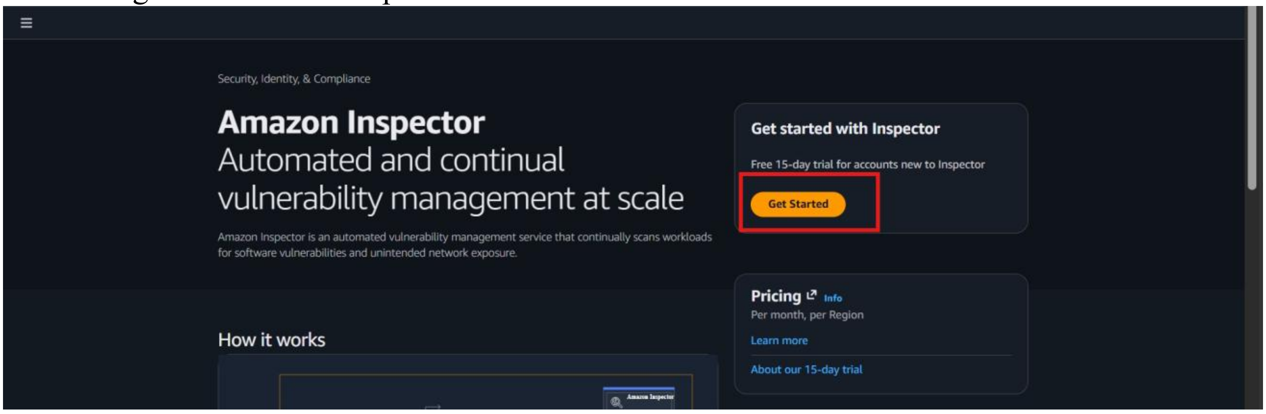
• Configure the following inbound rules:
RDP (TCP 3389) – Source: Anywhere (0.0.0.0/0)
HTTP (TCP 80) – Source: Anywhere (0.0.0.0/0)
Custom TCP (Port 21) – Source: Anywhere (0.0.0.0/0)



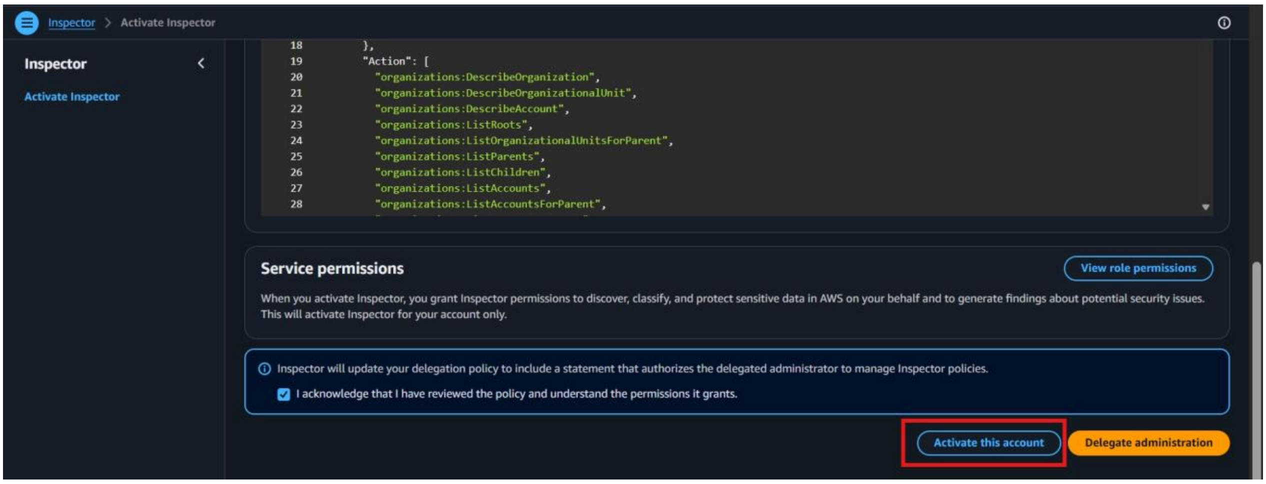
• Launch the EC2 instance and wait until its status changes to **Running**.
• Wait until the instance reaches the Running state.

Step 2 – Open Amazon Inspector

- Navigate to Amazon Inspector console. Click **Get Started**.



- If Amazon Inspector is not already enabled, review the permissions displayed on the activation page.
- Select the acknowledgment checkbox to grant the required permissions.
- Click Activate this account.

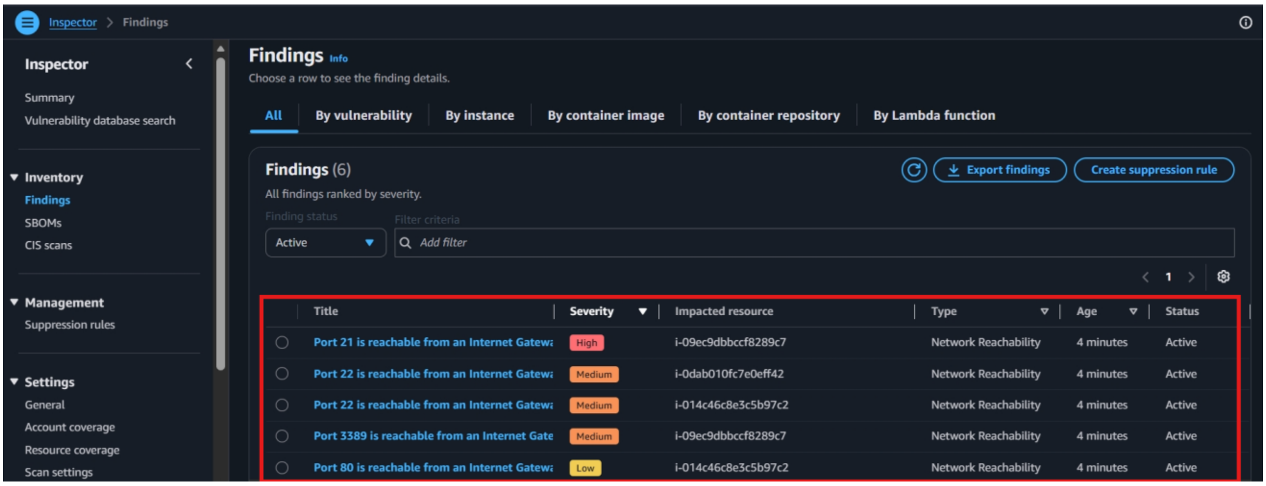


- Wait a few moments while Amazon Inspector is activated and begins scanning supported AWS resources.
- Verify that the dashboard displays the message:

  "Welcome to Inspector. Your first scan is underway."

Step 3 – View Amazon Inspector Findings

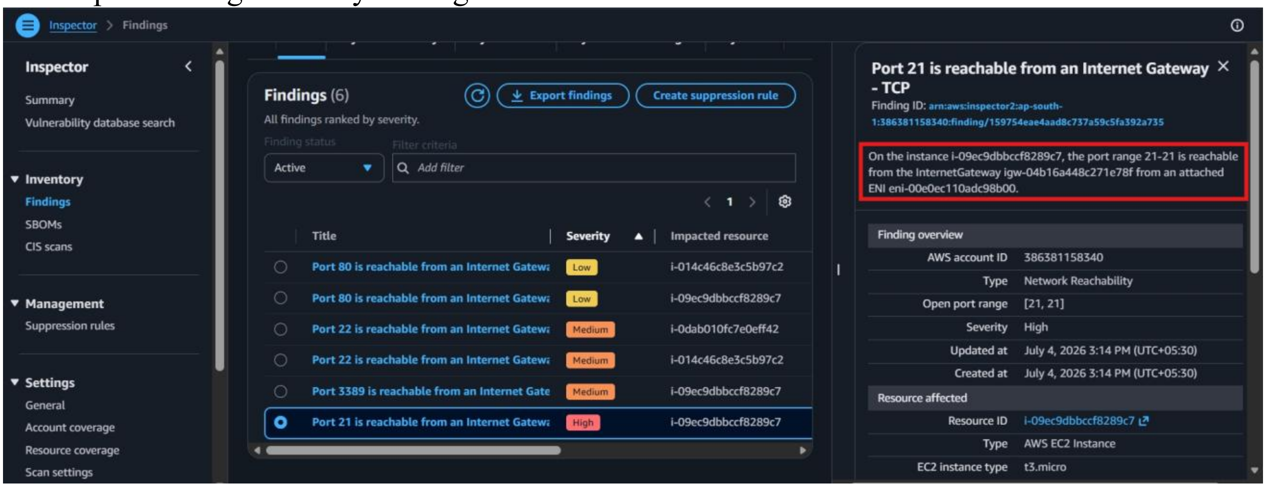
- In the Amazon Inspector console, select Findings from the left navigation pane.
- Wait for the initial scan to complete.



- Review the reported findings categorized as:
  - High
  - Medium
  - Low
  - Informational

Step 4 – Review the High Severity Finding

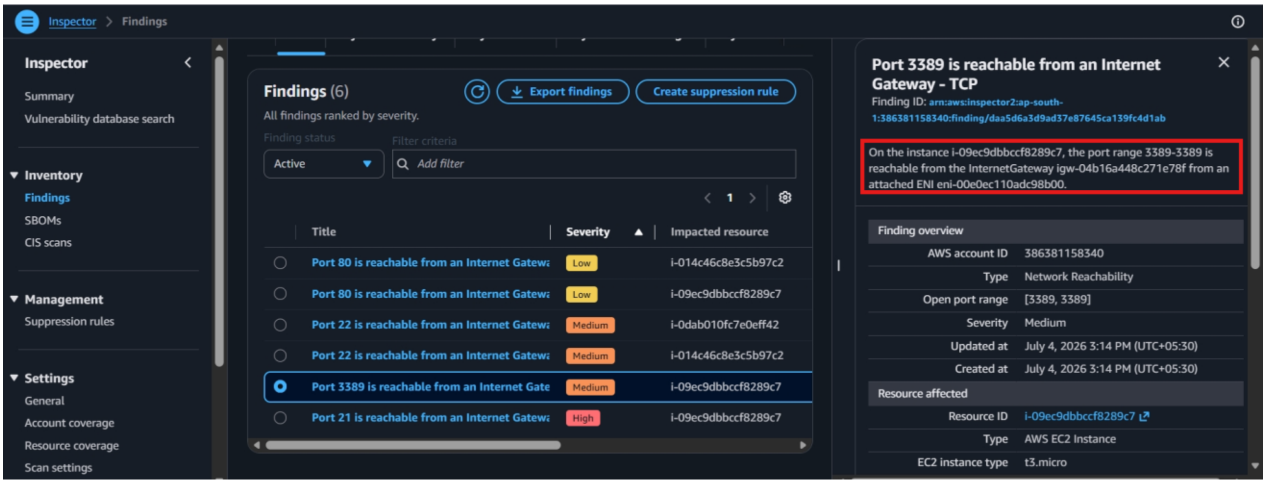
- Open the High severity finding.



- Verify that TCP Port 21 (FTP) is reported as publicly accessible from the internet.

Step 5 – Review the Medium Severity Finding

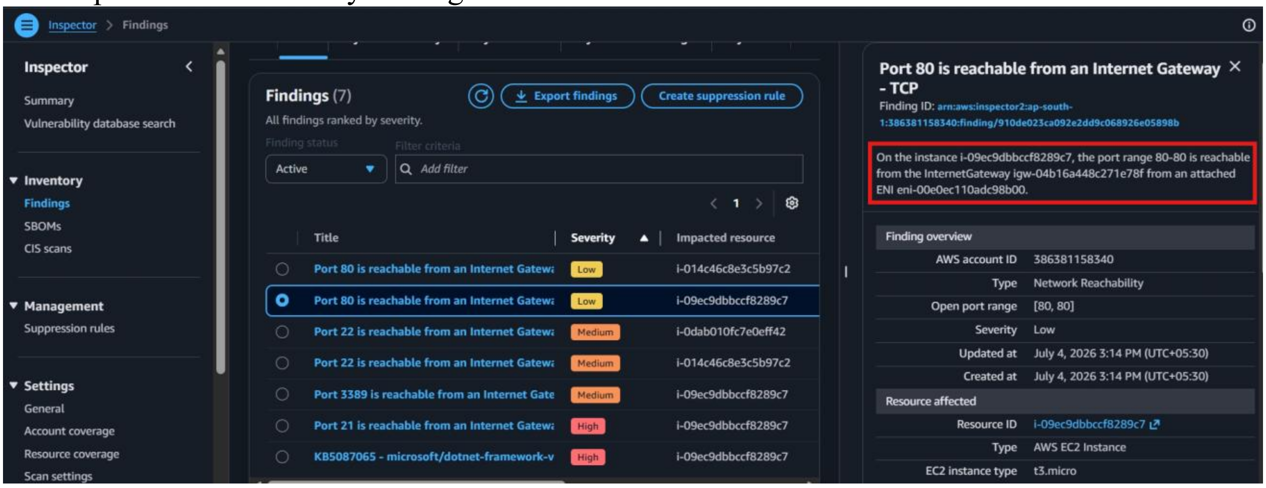
- Open the Medium severity finding.



• Verify that TCP Port 3389 (RDP) is reported as publicly accessible.

Step 6 – Review the Low Severity Finding

• Open the Low severity finding.



• Verify that TCP Port 80 (HTTP) is reported as publicly accessible.

Verification

• Amazon Inspector successfully analyzed the EC2 instance for network exposure.
• Publicly accessible ports were identified successfully.
• Security findings were categorized by severity.
• The assessment findings were reviewed to identify potential network security risks and exposed services.